24-25秋冬普物II(H)fmh期末

选择 5题

1.磁化强度矢量M的单位

2.两块偏振片摆放的让unpolarized的光无法通过，现在在中间摆一块和后一块偏振片夹角45°的偏振片，则透出的光是入射光强的（）倍

3.单缝衍射把绿光换成红光，如果条纹不变，则可能是调节了

A.调大了缝宽

B.调小了缝宽

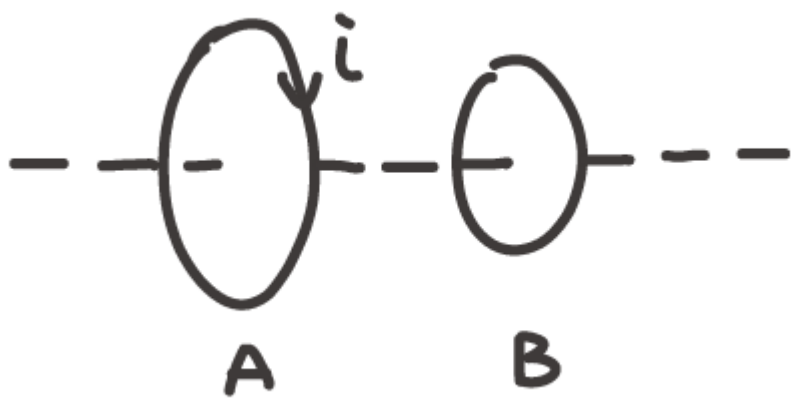
C.调大了缝屏距

D.调大了光强

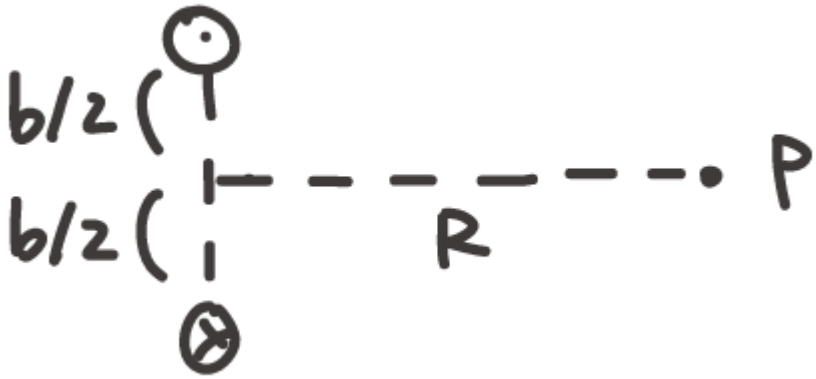
（还有两题太简单忘掉了）

填空 （39%）

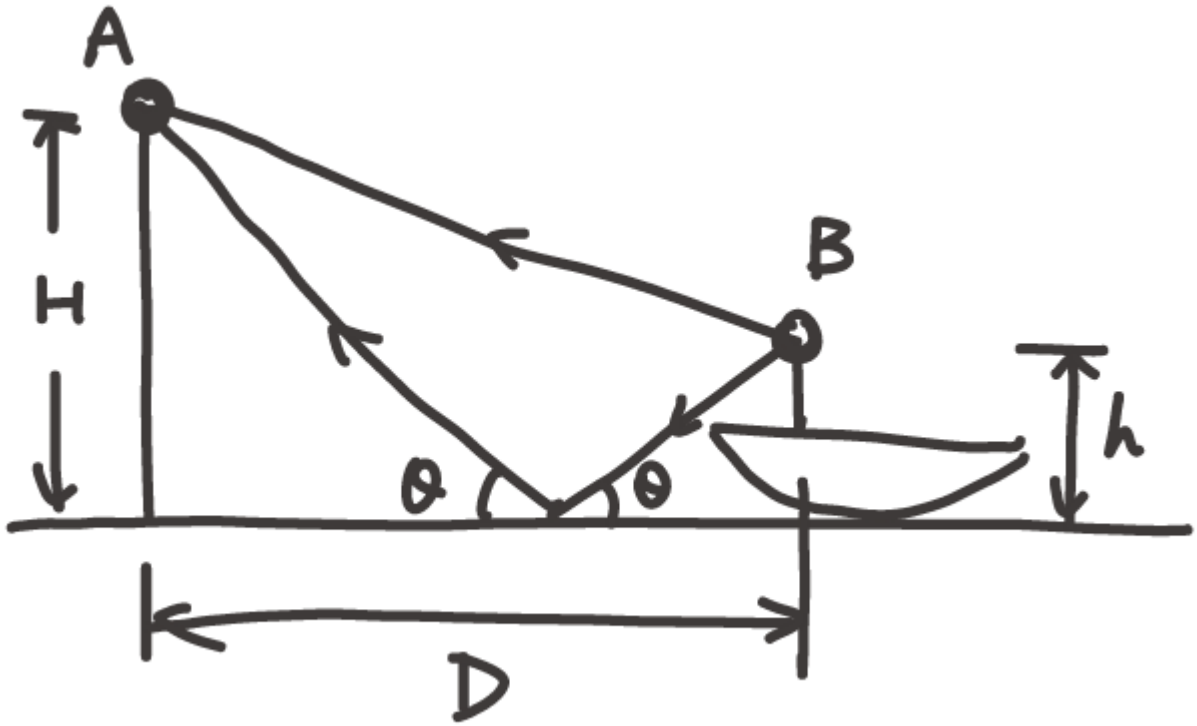
6.互感线圈其中一个突然施加电流，另一个的感生电流方向是_____，受力方向是_____



7.两根无限长直导线对P的磁场_____



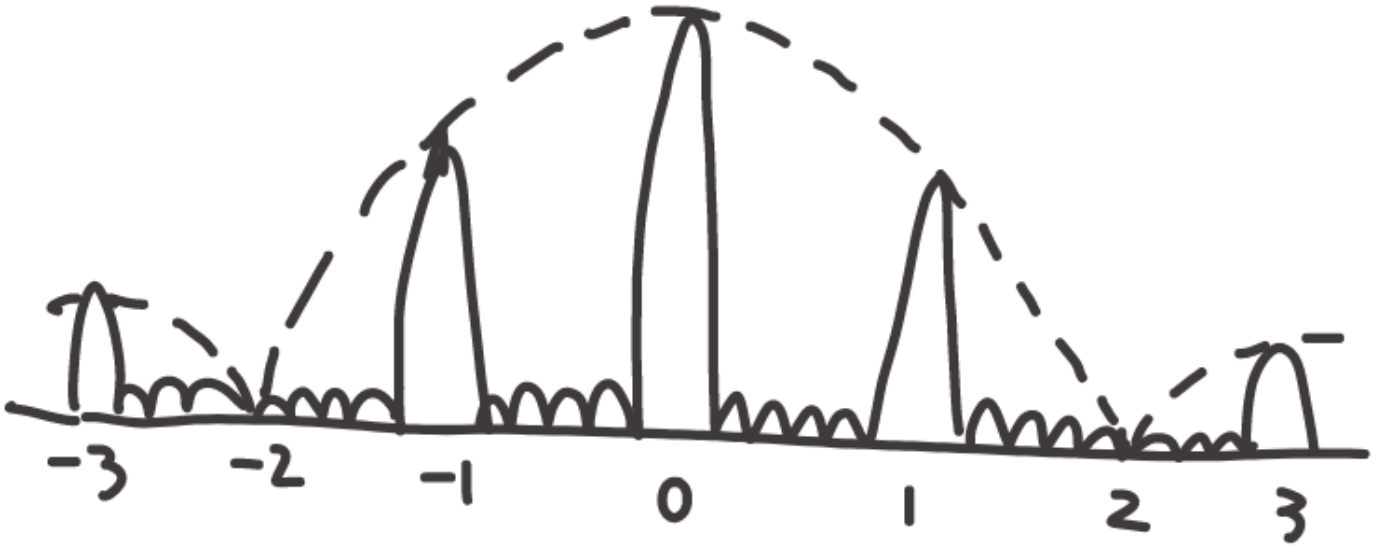
8.B发波长$\lambda = 3.43m$的信号，A接收。现在船从远处驶来，求第一次B无法接收到信号时相距的距离$D =$ _____.



干涉 解方程 小心半波损失

9.磨镜者公式默写_____

10.默写N条光栅的光强公式_____；给单缝衍射和光栅衍射的图（去年也考了），$N =$ _____，$\dfrac{d}{a} =$ _____

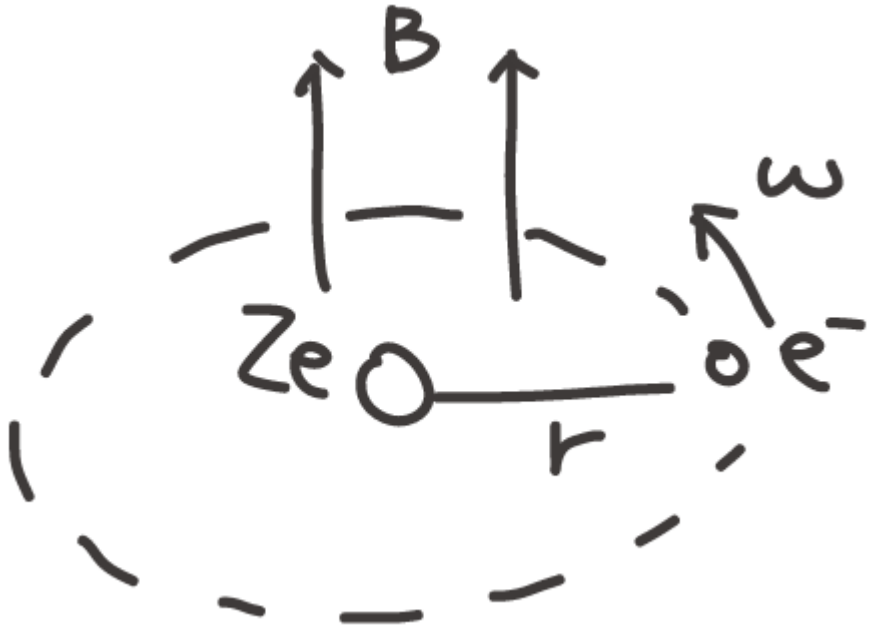


11.阳光照耀在Zi Jinggang的大地上，亮度$10mW/m^2$。光的E的峰值为_____，光对黑地（perfectly absorbed）的压强_____，$1m^2$太阳能板1sec吸收的能量为_____。

12.康普顿散射公式默写 $\Delta\lambda =$ _____

13.一维定态薛定谔方程默写
（去年考了一维时变薛定谔方程和动量、能量算子）

大题 （46%）
（10%）电子绕核运动，施加磁场B。



a.求变化的角频率$\Delta\omega$.
b.求$\Delta\omega$引起的磁矩变化$\Delta\mu$.

（10%）作业题

1. Figure 33-57 shows an arrangement known as a *Helmholtz coil*. It consists of two circular coaxial coils each of N turns and radius R, separated by a distance R. They carry equal currents i in the same direction. Find the magnetic field at P, midway between the coils.

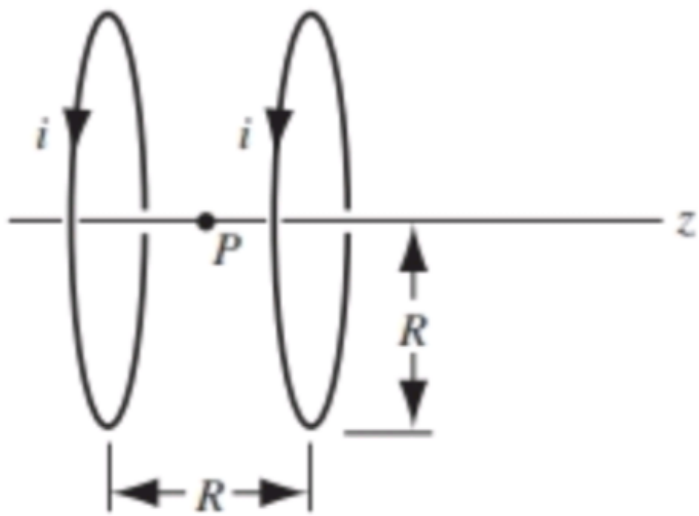
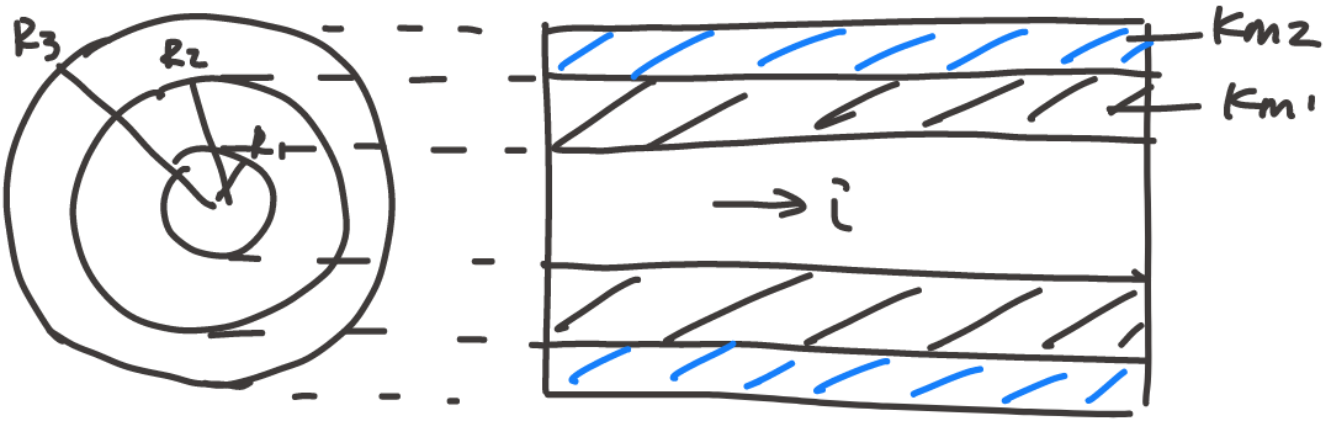


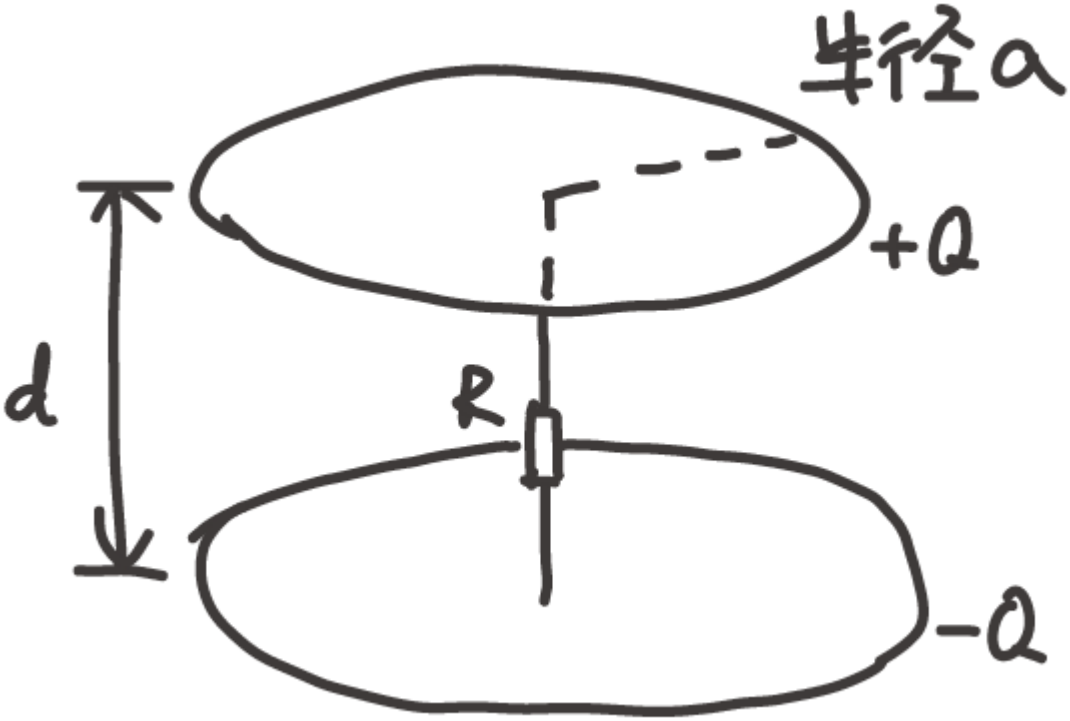
FIGURE 33-57. Problems 1, 3, and computer problem 1.

（13%）半径为R_1的无磁材料通电流i，外包磁导率κ_{m1}、κ_{m2}半径R_2、R_3的顺磁材料。



a.求顺磁材料中的磁场强度$\vec{H}$.
b.求半径$r = R_1, R_2, R_3$处的磁化强度矢量$\vec{M}$.

（13%）电阻R直接连接电容器。$t = 0$时突然给电容充Q_0的电荷。



a.求电容
b.求电压
c.求板间电场
d.求板间磁场B as a function of position and time
e.求Poyting vector S

☲ 2025.1.5